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SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)
CONTINUOUS ASSESSMENT EXAM-1

Programme: B.E/B.TECH
Course Code: SPHB1104
Course Name: PHYSICS
Batch : 2025 - 2029 SET-B

Max. Marks: 50
Time: 2 Hours
Sem: I.
Date: 17.02.2026

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Explain the term Elasticity	CO1
2.	A spanner of length 20cm is used to apply a force of 15N perpendicularly to it. Calculate the moment of the force.	CO1
3.	Differentiate uniform and non-uniform bending.	CO1
4.	(i) The unit cell with a=b not equal to c and all axis equal and all interfacial angle 90° a) cubic b) Monoclinic c) Tetragonal d) Triclinic (ii) Which of the following has the highest APF? a) SCC b) BCC c) FCC d) Diamond Cubic	CO1
5.	Enlist the major differences between Schottky and frenkel defect	CO2
6.	Calculate the inter planar spacing for (100) plane , if the lattice constant is 4A°	CO2
7.	Write note on edge dislocation	CO2

Part-B Answer ALL the questions (3 × 12 = 36)

8.	(i) State Hook's law? Give its mathematical relation (2+1 =3 mark) (ii) With proper diagram explain the different modulus of elasticity (3*3=9 mark)	CO1
	(OR)	
9.	Derive the depression at the free end of the cantilever subjected to a load at the free end.	CO1

10.	Experimentally evaluate the Young's modulus of a beam by subjected to non-uniform bending, by measuring the depression at the mid-point	CO1
	(OR)	
11.	Discuss briefly the classification of crystal system based on unit cell parameters with examples.(6+6 mark)	CO2

12.	Calculate the atomic packing factor for Simple cubic system (SCC) and Body centered cubic system (BCC) (6 *2=12mark)	CO2
	(OR)	
13.	Slow evaporation method for the synthesis of single crystal.	CO2